

# Gopal Kumar

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## EDUCATION

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University of California, Berkeley

August 2022 - May 2023

**Master of Engineering, FinTech (Industrial Engineering & Operations Research)**

Relevant Coursework: *optimization analytics; risk modeling, simulation & data analysis; applications of machine learning to electronic markets; introduction to financial engineering; financial engineering systems I*

Indian Institute of Technology, Patna

July 2014 - May 2018

**Bachelor of Technology, Computer Science & Engineering**

Relevant Coursework: *intro to computing; data structures & algorithms; probability theory; software engineering; databases*

## SKILLS

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**Scripts:** VBA, Python, R, AHK, Auto IT, AMPL (CPLEX)

**Web:** Bootstrap, HTML5, JavaScript/jQuery, CSS, Django, Selenium

**Databases:** MySQL, PostgreSQL, Oracle

**Tools:** MS Excel, MS PowerPoint, EViews, Tableau, JIRA, Git, AWS

**Data tools/software:** Haver Analytics, Bloomberg, Refinitiv, FactSet, JPM DataQuery, Snowflake

**Libraries:** pandas, NumPy, matplotlib, seaborn, scikit-learn, statsmodels, keras

**Interests:** macroeconomics, policy research, operations, data science, analytics, financial services, automation, time series analysis, alternative data, quantitative research.

**Languages:** English (Fluent), Hindi (Mother-tongue).

## WORK EXPERIENCE

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**JP Morgan**

**Mumbai, India**

**Associate, US Economic Research,** Corporate & Investment Banking

Jan 2020 – May 2022

- Covered US and Canadian economic data releases and policy reports. Also worked with India economic research team.
- Supported senior analysts in the team with modeling tasks that involved data cleaning, wrangling, modeling. Performed quantitative forecasting and created informative visualizations, which were later translated in daily, weekly, or monthly publications.
- Coauthored **20+ research notes** commenting on the data releases and economic policies. Avg. weekly views ~ 1000+ clients.
- Responsible for publishing **Canada DataWatch** (one pager summarizing weekly releases to be included in Global DataWatch).
- Published articles in top media publications like **Bloomberg Quint** and **Yahoo Finance** under the banner of JP Morgan.
- Secured **3rd position** in DataViz competition, analyzing Foursquare data for JP Morgan's Corporate & Investment Banking.
- Part of the Global Fixed-Income research team which topped rankings by institutional investors in 2020 and 2021.
- Consistently recognized with outstanding feedback in annual performance appraisals.
- References- Jesse Edgerton, Daniel Silver, Dan Weitzenfeld, Peter McCrory, Silvana Dimino, Sajjid Chinoy, Toshi Jain.

**JP Morgan**

**Mumbai, India**

**Analyst, FinTech & Automation,** Global Research, Corporate & Investment Banking

Jul 2018 – Dec 2019

- Executed multiple automation and analytics projects wherein the technology spectrum ranged from Python, R and VBA Macros to automation scripts such as AHK and Auto IT coupled with backend web frameworks like Django.
- Initiated, designed, and executed innovative projects on the periodical generation of research reports, web scraping, setting up databases, and building automation tools to support 1500+ analysts in JP Morgan research.
- Collaborated and saved approximately 4000+ man hours in a calendar year by optimizing processes and making them more efficient based on the inputs gathered from all the research analysts across Global research in JP Morgan.
- Managed the setting up of the team, being the first full-time recruit, improved the organizational structures between cross-functional teams, and enhanced communication with the clients. Facilitated expansion by participating in hiring processes.

## PUBLICATIONS

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Peer-Reviewed Publications

- **“Latent Dirichlet Allocation Modeling for CPS Patent Topic Discovery”**, with Usharani Hareesh Govindarajan & Amy J.C. Trappey in Proceedings of the 2018 International Conference on Industrial Enterprise and System Engineering (IcoIESE 2018).
- Implemented LDA model for topic modeling which identified relationships between terms and topics explaining underlying IP dynamics.

## Media Publications

- “**India: A Powell-ian deconstruction of India’s CPI**”, with Sajjid Z Chinoy & Toshi Jain in Bloomberg, Oct 2021, JP Morgan.  
- Research note assessing whether inflation was being driven by a few outliers or was more broad-based. We created three different inflation estimates for India’s CPI basket (~300 time series) – ‘diffusion index’, ‘trimmed mean index’ and ‘disproportionality index’.

## J.P.Morgan Markets Publications

- “**US: Big data batting 1-for-1**”, with Jesse Edgerton & Dan Weitzenfeld, Jun 2020, JP Morgan.  
- Research note explaining how we identified 20+ high-frequency alternative data series during the pandemic to assess its real-time impact. We designed and performed daily data aggregation of alt-data to complement traditional forecasting of economic data.
- “**US: Lessons from a year of alt data watching**”, with Jesse Edgerton & Dan Weitzenfeld, Mar 2021, JP Morgan.  
- Note summarizing utility of alternative data. Concluded high-frequency data provided early signals of dramatic turning points during the pandemic.
- “**India in 2022: Five themes to track**”, with Sajjid Z Chinoy & Toshi Jain, Jan 2022, JP Morgan.
- “**T&E rebounds from Delta but remains depressed**”, with Jesse Edgerton, Oct 2021.
- “**The unemployment experiment has begun**”, with Jesse Edgerton & Daniel Silver, June 2021.
- “**High-frequency data muddling through May**”, with Jesse Edgerton, May 2021.
- “**COVID effects are lingering in some states**”, with Jesse Edgerton, Apr 2021.
- “**Tailwinds converging on March retail sales**”, with Daniel Silver & Jesse Edgerton, Apr 2021.
- “**Stimulus adds fuel to spending rebound**”, with Jesse Edgerton & Daniel Silver, Mar 2021.
- “**Heat maps feel the Texas chill**”, with Daniel Silver & Jesse Edgerton, Feb 2021.
- “**The crowd noise is fake**”, with Jesse Edgerton, Oct 2020.
- “**Mixed news on swing state labor markets**”, with Jesse Edgerton, Aug 2020.
- “**Timely trackers diverge only slightly across states**”, with Jesse Edgerton, May 2020.
- “**Some timely trackers start to inch up**”, with Jesse Edgerton, Apr 2020.
- “**Travel trackers stay flat at low levels**”, with Jesse Edgerton, Apr 2020.
- “**Timely trackers hitting bottom in COVID-19-affected sectors**”, with Jesse Edgerton, Apr 2020.
- “**Timely travel trackers either falling or flat**”, with Jesse Edgerton, Apr 2020.

## PROJECTS

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- “**Synthetic data generation for improving time series forecasts**”, Jan-May 2023, UC Berkeley Capstone project. (Tech- Python)  
- Enhancing long-term returns in stock market through the application of data augmentation techniques in neural networks.
- “**US: Jobs tracker based on alternative data**”, 2020-21, JP Morgan. (Tech- Python)  
- Reimagined the ‘payroll jobs tracking’ by utilizing high-frequency alternative data series (~25 series) to forecast jobs growth.
- “**US: Retail sales tracker based on Chase card consumer data**”, 2020-2021, JP Morgan. (Tech- Python)  
- Built a nowcasting tracker for monthly retail sales using proprietary Chase card consumer spending data.
- “**PDF Tagger and sentiment analysis using NLP**”, 2019, JP Morgan. (Tech- Django, Python, JavaScript)  
- Designed the backend of the PDF tagging system on Django. Created the pipeline where PDF documents were tagged in real time.
- “**JP Morgan Research News Aggregator**”, 2018-2019, JP Morgan. (Tech- Django, Python, HTML, CSS, JavaScript)  
- Designed, built, and deployed a news aggregator system for individual research analysts using RSS feeds, News API, web scrapers.
- “**Microblog retrieval in disaster situations (IR models)**”, IIT Patna, 2017-18. (Tech- Python: nltk, gensim, textblob, scikit-learn)  
- Designed and evaluated utilities of various systems that retrieve information from tweets during disasters pertaining.
- “**Trans-OTP Authentication**”, IIT Gandhinagar, 2016. (Tech- PHP, MySQL, HTML, CSS)  
- Studied two-factor authentication systems and implemented a novel way of authenticating users in Online Banking System.
- “**Digital marketing strategy using social networks**”, Summer Research Project, NTHU (Taiwan), July 2017  
- Researched and evolved a digital marketing strategy for OBM centric development for Taiwan’s high-tech-service industry.

## LEADERSHIP

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- **Affinity Group Leader, FinTech** – Fung Institute, University of California, Berkeley, (Aug 2022 –Present), Berkeley, CA.  
- Connecting and networking with industry experts and alums to explore, learn and participate in FinTech events and related domains.
- **Sponsorship & Events Coordinator, Annual Techno-Cultural Student Fest** – IIT Patna, (July 2016 – May 2017), Patna, India.  
- Worked on new strategies for acquiring desired sponsorship. Managed organization of 30+ events in a span of three days.

## AWARDS

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- Awarded the Fung Fellowship at **Fung Institute for Engineering leadership** fellowship, UC Berkeley (2022), United States.
- Awarded National Tsing Hua University’s GEL (**Global Engineering Leadership**) fellowship for research (2017), Taiwan